IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

MATHEMATICS 1B

Tuesday, 29 April 2025, 2:30 PM

Time allowed: 2:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks (25% each).

NO CALCULATORS ALLOWED. Formula sheet will be provided.

Any special instructions for invigilators and information for candidates are on page 1.

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THE QUESTIONS

1. a) Three vectors in $\mathbb{R}^3$ are given as $\underline{\mathbf{a}} = (0,1,3)^T$, $\underline{\mathbf{b}} = (1,2,4)^T$, and $\underline{\mathbf{c}} = (1,-1,0)^T$.

(i) A plane P_1 through the origin is obtained by taking all linear combinations of $\underline{\mathbf{a}}$ and $\underline{\mathbf{b}}$. A plane P_2 is obtained by adding $\underline{\mathbf{c}}$ to every point in P_1. Obtain the cartesian equation of plane P_2. [4]

(ii) Use projection to find the distance from the point $M(3,-5,-2)$ to the plane P_2. Find the vector equation of the line L_1 through M, orthogonal to P_2. Finally, obtain the point of intersection N of this line with P_2, and the magnitude of the vector $\overrightarrow{MN}$. Comment on this! [5]

b) A double integral is given as

$$I = \int_{x=0}^{\pi/2} \int_{y=x}^{\pi/2} \cos\left(\frac{2y^2}{\pi}\right) \, dy \, dx \, .$$

To evaluate this integral, first sketch the region of integration R, then change the order of integration, and finally obtain the value of I. [6]

c) A function of two variables is given as

$$f(x,y) = (x^2 - 1 + y)(x^2 - 1 - y) \, .$$

(i) Obtain all stationary points of $f(x,y)$ and use the Hessian determinant to classify them as maximum, minimum or saddle point. [5]

(ii) Sketch the zero contour of $f(x,y)$. Separately, use the zero contour and the information from (i), to sketch the full contour plot. Be sure to explain your reasoning. [5]

2. a) Obtain the solution of the first-order Ordinary Differential Equation

$$x\sin t - t - x^2 e^{2t} \;=\; \left[\cos t + \sin x + x e^{2t}\,\right] \frac{dx}{dt}$$

which satifisies the initial condition $x(0) = 1$. [6]

b) Let a function be defined as $f(x) = \begin{cases} 0 & \text{for} \quad 0 \le x < \pi/2\,, \\ \pi - x & \text{for} \quad \pi/2 \le x < \pi\,. \end{cases}$

Let $g(x)$ be the even extension, $h(x)$ the odd extension of $f(x)$, both with period $T = 2\pi$.

(i) Sketch $g(x)$ and $h(x)$, for $-3\pi \le x < 3\pi$. If Fourier series for $g(x)$ and $h(x)$ are calculated (do not do this yet!) what is the value each Fourier series takes at $x = -3\pi/2$? State the rate of decay of the Fourier coefficients for each series. [6]

(ii) Now obtain the first six non-zero terms of the Fourier series for $h(x)$. [8]

(iii) Use an appropriate choice of x in (ii), to find $\displaystyle\sum_{n=1}^{\infty} \frac{1}{(2n-1)^2}$ [5]

3. a) A real-valued signal $f(t)$ can be written as the sum of its even and odd parts: $f(t) = f_e(t) + f_o(t)$, where $f_e(t)$ and $f_o(t)$ are even and odd functions, respectively. If $F(\omega)$ is the Fourier transform of $f(t)$, obtain the real and imaginary parts of $F(\omega)$ in terms of the Fourier transforms of $f_e(t)$ and $f_o(t)$, explaining your reasoning. [5]

b) Obtain $F(\omega)$, the Fourier transform of $f(t) = tH(t+1) - tH(t-1)$, where $H(t)$ is the Heaviside function, and give $F(\omega)$ in terms of trigonometric functions. [5]

c) (i) Use the definition of the Laplace transform to prove the second shift theorem: if $\mathscr{L}[f(t)] = \bar{f}(s)$, and $H(t)$ is the Heaviside function, then

$$\mathscr{L}[f(t-a)H(t-a)] = e^{-sa}\bar{f}(s) .$$

State any necessary condition. [4]

(ii) Use convolution and the second shift theorem to obtain the following inverse Laplace transform:

$$\mathscr{L}^{-1}\left[\frac{e^{-2s}}{s(s^2+1)}\right] . \qquad [6]$$

(iii) Use Laplace transforms and (ii), to solve the Ordinary Differential Equation

$$\frac{d^2x}{dt^2} + x = \begin{cases} 0 & \text{for} \quad 0 \le t < 2 , \\ 1 & \text{for} \quad t \ge 2 , \end{cases}$$

which satisfies the initial conditions $x(0) = x'(0) = 1$. [5]

4. a) Two matrices are given as

$$A = \begin{pmatrix} 1 & 2 & 3 \\ -1 & 1 & 1 \\ \phi & 0 & 1 \end{pmatrix}, \quad \text{and} \quad B = \begin{pmatrix} 1 & 3 & 2 \\ -1 & 1 & 1 \\ 2 & 1 & -1 \end{pmatrix}$$

(i) Let $\phi = 2$. Find $\det(A)$, as well as $\det(B)$ and $\det(AB)$. Finally, use Gauss-Jordan elimination to obtain A^{-1}. [6]

(ii) Let $\phi = 3$. Give conditions on $\underline{\mathbf{b}} = (1, \theta, 2)^T$ so that the matrix equation $A\underline{\mathbf{x}} = \underline{\mathbf{b}}$ has no solutions, or infinitely many solutions. Can the system have a unique solution? Either way, explain why. Finally, for the case of infinitely many solutions, find these and give a geometric interpretation. [6]

(iii) Use row operations to obtain the echelon form of B, the upper-triangular matrix U. The $LU-$decomposition of B requires the lower-triangular matrix L, so that $LU = B$. With no further calculations needed, write down L. [4]

b) Let a matrix be given as

$$A = \begin{pmatrix} -1 & 1 & 1 & 2 \\ 0 & 1 & 2 & 3 \\ 2 & 1 & 4 & 5 \\ 0 & -2 & -4 & -6 \end{pmatrix}.$$

Use Gaussian elimination to obtain the echelon form of A, identify rank, pivot and free columns, and hence find bases for nullspace $N(A)$ and column space $C(A)$. What does the basis for the nullspace tell us about the columns of A? Finally, obtain a particular solution of $A\underline{\mathbf{x}} = \underline{\mathbf{b}} = (7, 4, -2, -8)^T$, and with no further calculation, write down the general solution. [9]